Unit 1 – Using Objects and Methods Name: _____
Checkpoint Practice 3

1) Write what the following code segments will evaluate to.

```
String str = "Let's practice some";
```

Code	Evaluates To
`str.substring(6, 14)`	
`str.substring(0,2) + str.substring(15)`	
`str.indexOf(" ")`	
`str.substring(16).length()`	
`str.substring(str.length() - 7)`	
`str.substring(str.substring(2, 11).length())`	

2) Write what the following code segments will evaluate to.

```
String phrase = "hello world!";
```

Code	Evaluates To
`phrase.substring(2)`	
`phrase.substring(phrase.length()-2)`	
`phrase.substring(0, 5) + phrase.substring(6)`	
`phrase.substring(phrase.length()-4, phrase.length()-3)`	

3) Write what the following code segments will evaluate to.

```
int x = 5;
double y = 2.5;
```

Code	Evaluates To
`x + Math.pow(x, 2) % 3 + (int) y`	
`(int)(Math.pow(y, 2) + 0.5) % 4`	
`x / y + Math.abs(y - x) + (int) Math.sqrt(x * 4)`	
`Math.abs(y - Math.pow(Math.sqrt(16), 2))`	

4) Write a Java expression to find $\left(\sqrt{64}+3\right)^2$

5) Write an expression that calculates the absolute value of −15 then adds it to the square root of 121.

6) Write an expression that generates a random integer from 10 to 100.

7) Write an expression that finds the absolute difference between 12 and 25 raised to the power of 2.

8) Write an expression that takes the square root of 36 and multiplies it by a random integer between 1 and 5.

9) For the following code segments, determine if a positive number, negative number, or 0 is printed and explain why.

Code Segment	Number Printed and Why

`String word1 = "Apple";`
`String word2 = "Banana";`
`System.out.println(word1.compareTo(word2));`	

`String word1 = "Zebra";`
`String word2 = "Monkey";`
`System.out.println(word1.compareTo(word2));`	

`String word1 = "Hello";`
`String word2 = "Hello";`
`System.out.println(word1.compareTo(word2));`	

`String word1 = "hello";`
`String word2 = "Hello";`
`System.out.println(word1.compareTo(word2));`	

`String word1 = "Program";`
`String word2 = "Programming";`
`System.out.println(word1.compareTo(word2));`	

_____ 10) Consider the following method, which is intended to return true if at least one of the three strings `s1`, `s2`, or `s3`, contains the substring "no", otherwise the method should return false.

```
public boolean containsNo(String s1, String s2, String s3)
{
     String together = s1 + s2 + s3;
     return (together.indexOf("no") != -1);
}
```

Which of the following method calls (located in the same class) demonstrates that the method does **not** work as intended?

(A) `containsNo("note", "ignore", "nomad");`
(B) `containsNo("laughing", "happy", "noteworthy");`
(C) `containsNo("solution", "over", "loving");`
(D) `containsNo("enough", "button", "octopus");`

_____ 11) Which of the following correctly gives random integers between –7 and 2 inclusive?

(A) `int n = (int)(Math.random() * 8) - 7;`
(B) `int n = (int)(Math.random() * 9) - 7;`
(C) `int n = (int)(Math.random() * 10) - 7;`
(D) `int n = (int)(Math.random() * 11) - 7;`

_____ 12) Consider the following code:

```
String word1 = "butter";
String word2 = "letter";
System.out.println(word1.indexOf(word2.substring(3)));
```

What is the output?

(A) –1
(B) 1
(C) 2
(D) 3

_____ 13) If the following returns a positive number, what must be true about `str1` and `str2`?

```
str1.compareTo(str2)
```

(A) `str1` comes before `str2` alphabetically
(B) `str1` is longer than `str2`
(C) `str1` equals `str2`
(D) `str1` comes after `str2` alphabetically